Name: Mamadou Cellou Diallo, May Liu, Richard Nguyen
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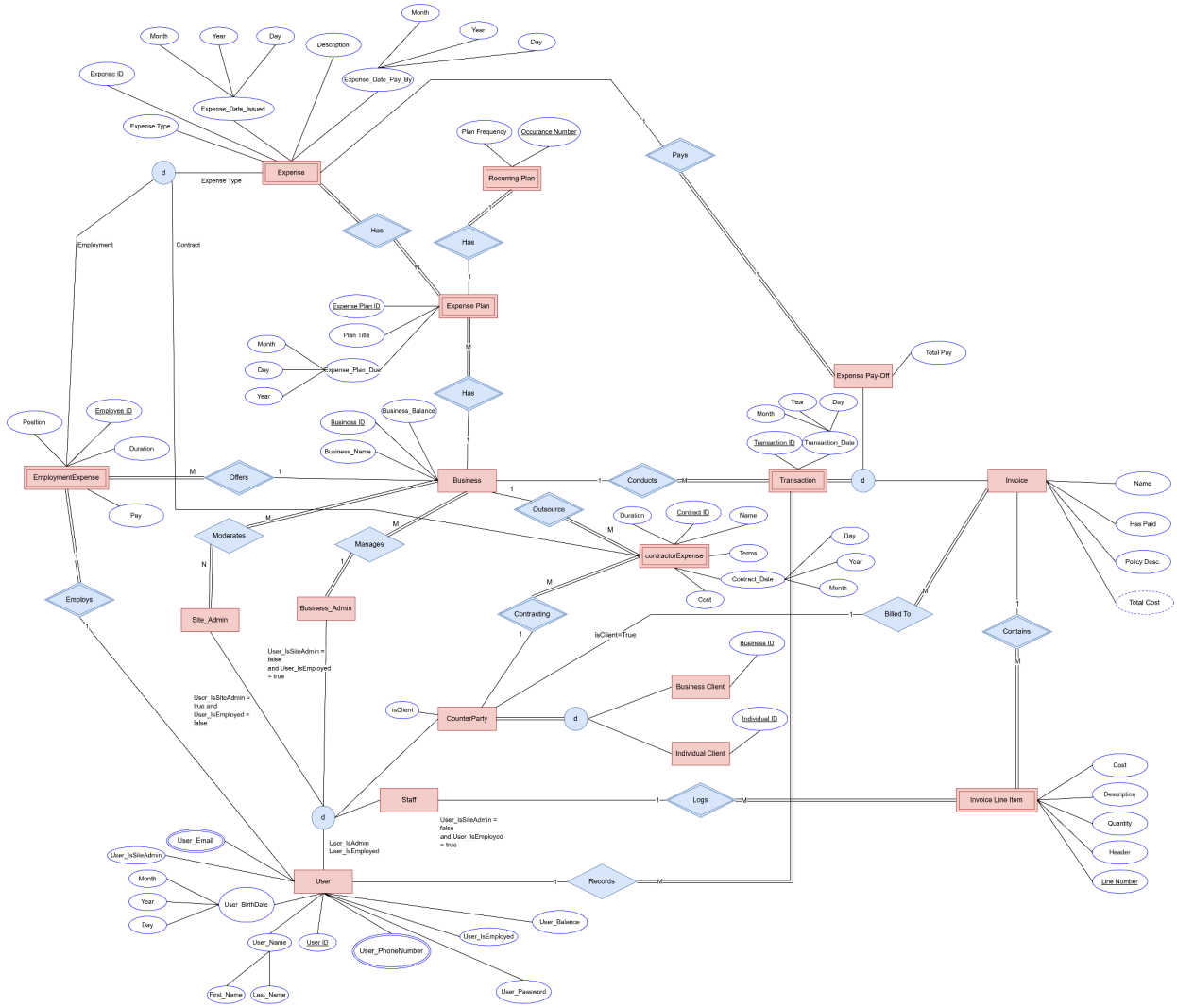
RM Progress Report

Note: New and Scaled-Down EERD Diagram, which will consequently affect the RM Diagram



https://drive.google.com/file/d/1hn31_XwWazhPIkyaMBHT5gRwrdbvyqOi/view
(Link to EERD Diagram in case the picture does not load or is too blurry)

Assumptions for Revised EERD Diagram:

- Business is modelled as a single entity for now.
- A user may not belong to a single business. They may be unemployed.
- Each invoice contains one or more line items, and line items cannot exist independently. Line items are logged by staff members of a business.

- Invoice totals are derived from line items (sum of line totals).
- Client payments are applied to exactly one invoice. And we only model full payments for now.
- The system stores past expense plans by businesses, so they are retrievable from the database.
- Expenses can be highly specialized using a type attribute; however, if they are of the type contract or expense, additional specific attributes are added, allowing for greater detail and transparency when dealing with staff or businesses.
- Expense pay-offs represent actual cash outflows. Expense pay-offs must pay off exactly one expense in one expense plan.
- The expense entity can have a recurring cost plan with a number of occurrences and a plan frequency. This allows for dynamic due date generation/storage.
- Employments are stored in the database, including pay and duration, so we can calculate the total expense (which includes employment). It also includes the Employee ID, allowing employees to be identified in a business database.
- A Contract is defined as the business hiring other businesses/individuals to provide services, making it reliant on both parties to exist.
- There are 2 types of Admin: Web Admin and Business Admin. Business Admin manages their own businesses, while the Web Admin is responsible for managing the whole website, including moderating businesses. The Web Admin cannot be employed by a business because this would be a Conflict of Interest.
- End clients are known as CounterParty in the EERD.

https://drive.google.com/file/d/1FfC8CZZIzfGA7KjXOo3Poqe4qaIrr3NV/view?usp=sharing
(Link to RM Diagram in case the picture does not load or is too blurry)

Foreign Key Assumptions Made:

- Weak entities have their owners' PK as FK to model the dependence. For example expense plan has an FK Business ID, which is also a PK in this context, as an expense plan cannot exist without a business.
- EmploymentExpense requires the Users' PK, Business PK, Expense PK, and Expense Plan PK to track employment and its associated costs. It needs a user to be attached to, a business to hire that user, and an expense to be attached to an expense plan so that the business can pay for their work.

- The Contract entity has PKs userID and couterptID, which are both foreign keys that identify the user-business pair involved in the contract relationship. A contract needs two parties: the business and the client/CounterParty. Without it, a contract cannot exist.
- The Expense needs to be stored in an Expense Plan; therefore, the Expense requires an FK to the Expense Plan ID to form a full primary key with the Expense Plan.
- An Invoice Line Item needs to be logged by Staff and must be in an Invoice Bill; therefore, it needs an FK to the Staff (User ID) and an FK to the Invoice it will be found in.
- A Transaction requires a Business to actually conduct the transaction; therefore, a Transaction needs a Business ID.
- A Recurring Plan would allow us to reuse an Expense Plan for a future bill; however, it must be tied to an Expense Plan first, so it needs the Expense Plan's ID as the FK and the Business ID's FK through inheritance, since the Expense Plan is a weak entity.

Private Key Assumptions Made:

- Users have their own private key, called User ID, to distinguish them from other users.
- If a user is employed by a business, it has an Employee ID to help uniquely identify them in the business
- Businesses have their own private key in order to uniquely distinguish themselves in the database.
- Counterpart uses its own private keys to identify end-users, and each specialization also has its own key to uniquely identify sub-clients within a contract.